31.03.2026 Mentoring Session - Wenbo Yuan

#L2MAEL #Mentoring

OVERVIEW:

1. Tried Question One from the 91261 2025 EOT Paper (except final question)
2. Learnt how to interpret exam-styled questions
3. Learnt how to solve an Excellence-level question that uses the formula: $V = A \times r^t$ (where V stands for value, A stands for the initial value, r for rate (in percentage), and t for time)

Areas for improvement:

- Interpreting worded-questions that are written in English
- Knowing that taking the logarithm of both sides of the equation will help solve a problem

$$2^x = 4 \implies x \log 2 = \log 4 \implies x = \frac{\log 4}{\log 2}$$

- Using the formula $V = A \times r^t$ in solving depreciation/appreciation questions

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HOMEWORK:

1. (continue) Solve the equation:

$$9^{2x+3} = \left(\frac{1}{27}\right)^x$$

   Using logarithms (Hint was given during the session – refer to that hint).
2. Find the possible values of d if **real solutions exist** for $x^2 + 5x - 1 - d(x^2 + 1) = 0$.
3. There is **only one real solution** for the equation $(x + 3) - 4\sqrt{(x+3)} - 5 = 0$. Find that solution. HINT: let $x + 3 = w$
4. Peta spends $19 800 on equipment to set up a small office business. Each year, the equipment is expected to continuously depreciate by 17% of its previous year's value.

   (i) After how many years will the equipment first be worth less than $12 000? **(You have solved this in today's mentoring session. Please use the answer from today, and continue with (ii).**

   (ii) Jay starts a garden-care business at the same time as Peta. He paid $26 000 for his equipment which is expected to depreciate at 22% of its previous year's value. How long would it be until the equipment for each year of do their businesses have the same value?

Please visit https://nzqa.toasting.me/browse?search=91261 for more questions on this topic.